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Paper Code : PCC-CS402 Computer Architecture

UPID : 004442

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) The process of converting a program written in a high-level language into machine code is known as _____.
- (II) The process of copying data from main memory to cache memory is known as _____.
- (III) In _____ processors, the entire vector is processed in a single instruction cycle.
- (IV) In a centralized shared-memory architecture, _____ is used to ensure that multiple processors do not attempt to access the same memory location simultaneously.
- (V) In pipelining, a _____ hazard occurs when an instruction depends on the result of a previous instruction that has not yet completed.
- (VI) The _____ property of cache memory refers to the fact that when a block is placed in a cache, all the blocks from higher levels of the memory hierarchy that contain that block are also present in the cache.
- (VII) What are the basic techniques for increasing ILP?
- (VIII) What is the main emphasis of dataflow architecture?
- (IX) What is pipelining?
- (X) What is virtual memory?
- (XI) What are vector processors used for?
- (XII) What is the purpose of a systolic architecture?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. What is the purpose of virtual memory in computer architecture? [5]
3. What is the purpose of loop unrolling ? How does it increase instruction-level parallelism (ILP) in processor architectures? [5]
4. Explain the concept of a systolic array and give an example of an application that could benefit from this architecture. [5]
5. Explain the concept of branch prediction in computer architecture. What are the different techniques used for branch prediction? [5]
6. What is cache coherence? Explain its importance in multiprocessor systems. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. Compare and contrast the working of Direct Mapped, Set Associative and Fully Associative cache mapping techniques, highlighting their advantages and disadvantages. [15]
8. What are vector processors and how are they used in modern computing applications? [15]
9. What is distributed shared-memory architecture and how does it differ from centralized shared-memory architecture? <https://www.makaut.com> [15]
10. (a) Explain the concept of exception handling in computer architecture. [4]
 (b) What are the different types of exceptions that can occur? [4]
 (c) How can exceptions be handled efficiently in a pipelined processor? [7]
11. Consider a processor architecture that implements both pipelining and out-of-order execution. Assume that the processor has four functional units and a pipeline depth of eight stages. Each functional unit has a latency of two clock cycles and the processor can issue up to four instructions per clock cycle. Assume that the processor uses a branch predictor with a 90% accuracy rate.
 (a) What is the maximum instruction-level parallelism (ILP) that can be achieved by this processor? [8]

- (b) Assume that the program being executed contains a loop with a 10% taken branch. How many cycles are required to execute the loop, assuming that it contains 200 instructions and that the branch is taken every 10 iterations? [4]
- (c) What is the impact of the branch on the performance of the processor and how could this be mitigated? [3]

*** END OF PAPER ***